Shreya Nasa

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PhD student in theoretical computer science working on **coding theory** (interactive communication, error resilience under bit-flip and erasure noise), **sublinear algorithms and property testing**, and **extremal combinatorics** (hypergraph containers, Shannon capacity of graphs). Three years at Goldman Sachs on a large-scale knowledge graph and its data pipelines.

Education

MS + PhD in Computer Science — **Purdue University** Aug 2022–May 2028 (expected) GPA: 4.0/4.0

B.Tech. in ECE with Honors — **IIT Roorkee** Aug 2015–May 2019 CGPA: 9.13/10

Research Experience

Graduate Researcher — **Purdue University** Mar 2023–Present

Advisors: Prof. Elena Grigorescu and Prof. Kent Quanrud

- **Sublinear algorithms / property testing:** Used hypergraph containers to show that the independent-set property in q-uniform hypergraphs can be tested with $\widetilde{O}\big(q\rho^{2q-3}/(\varepsilon^2((q-2)!)^2)\big)$ samples, optimal in ε and exponentially better in q (RANDOM 2026).
- **Coding theory with feedback and erasures:** Characterized error resilience and maximal asymptotic communication rate with perfect feedback under bit flips and erasures (ISIT 2025).
- **Interactive error-correcting codes:** Designed adversarial attack strategies giving an upper bound of 7/24 on tolerable error fraction with separate corruption budgets (Allerton 2025).
- **Extremal graph theory:** Working on Shannon (zero-error) capacity $\Theta(G)$ of graphs.

Publications

Testing the Independent Set Property in Hypergraphs — *RANDOM 2026*

*Elena Grigorescu, **Shreya Nasa**, Cameron Seth**

Communication with Perfect Feedback for Bit Flips and Erasures — *ISIT 2025*

*Elena Grigorescu, **Shreya Nasa**, Maoyuan Song**

An Upper Bound for Interactive ECCs with Separate Budgets — *Allerton 2025*

***Shreya Nasa**, Maoyuan Song**

* Authors listed alphabetically by last name; all contributed equally. I presented all three papers at the respective conferences.

Teaching Experience

Graduate Teaching Assistant — **Purdue University** 2024–Present

- Head TA, graduate CS 58000: Algorithm Design (Fall 2024): led design and grading of all assignments and exams.
- TA, graduate CS 57800: Statistical Machine Learning (Spring 2024, Fall 2026) and CS 18000: Problem Solving and Object-Oriented Programming (Spring 2026).

Industry Experience

Associate / Analyst — **Goldman Sachs** Jun 2019–Jun 2022 Bengaluru, India

- Managed and extended the firm's HBase knowledge graph for Compliance's Relationship Analytics and Engineering team; integrated firm-wide e-communication data and graph-derived attributes with infrastructure and inference teams.
- Optimized MapReduce jobs in the communication-data pipeline, cutting runtime 3× (24 hours to 8 hours).
- Built a neural relationship-strength model between graph vertices, raising F1 from 0.40 to 0.89; onboarded and mentored junior teammates.

Skills & Interests

Theory: coding theory, interactive communication, property testing and sublinear algorithms, extremal combinatorics, graph theory, machine learning.

Programming: Python, C/C++, Java, SQL, MATLAB; PyTorch, TensorFlow, scikit-learn; MapReduce, HBase, Git.

Selected Achievements

Google Code Jam to I/O for Women: ranked 9th (2019) and 69th (2018).

Academic awards, IIT Roorkee: Prof. Ranvir Singh Shastry Cash Prize (second-highest CGPA among B.Tech. women); Rai Singh Jain Cash Prize (highest CGPA among B.Tech. women in CSE/ECE/Electrical, 2016 and 2017–18).